Project 3 Report: Assess Learners

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***Abstract*—This report examines the effects of leaf size on overfitting within decision tree models. Specifically, it investigates the decision tree learner (DTLearner) and the random tree learner (RTLearner), using the Istanbul.csv dataset provided. The key findings indicate that while overfitting can be influenced by leaf size, the implementation of bagging can mitigate this tendency. Moreover, a comparative analysis of DTLearner and RTLearner shows different performance metrics, offering insights into the superiority of one model over the other in specific contexts.

Introduction

Machine learning models, such as decision trees, can sometimes get too focused on the details of the data they're trained on and miss the bigger picture. This is like memorizing the answers to a practice test without understanding the subject, which means they don't do well when faced with new questions. This study looks at how much decision trees face this issue. It also sees if a technique called bagging helps them get better at taking new tests, and compares two types of decision trees (DTLearner and RTLearner) to see which one works better. The hypothesis is that increasing leaf size will initially reduce overfitting but may lead to underfitting at larger sizes, and that bagging will further reduce overfitting across various leaf sizes.

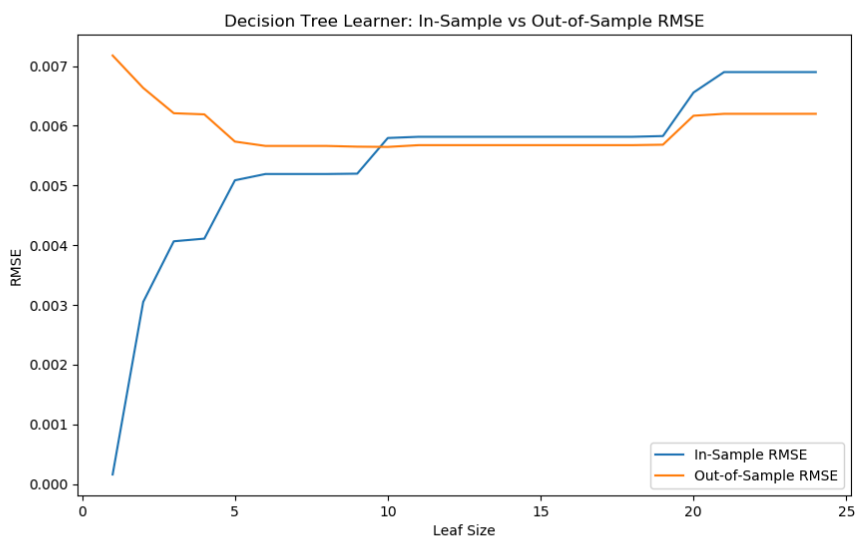
Methods

The experiments use the Istanbul.csv dataset, with the DTLearner model as the primary focus. The effect of leaf sizes on in-sample and out-of-sample Root Mean Squared Error (RMSE) was observed on decision tree models. Bagging was used in the second experiment to examine its impact on overfitting. The third experiment compared DTLearner and RTLearner using Mean Absolute Error (MAE) and Coefficient of Determination (R^2).

Discussion

1 EXPERIMENT 1: OVERFITTING ANALYSIS

Overfitting tends to occur in decision tree models when the model becomes too complex and starts to capture the noise in the training data. This is important because it detracts from its ability to generalize data it has not yet been introduced to, which can be a serious issue. This is typically seen when a model performs well on training data (in-sample) but poorly on testing data (out-of-sample).



The provided graph from Experiment 1 shows the in-sample versus out-of-sample RMSE for a DTLearner using the Istanbul.csv dataset with varying leaf sizes. Overfitting with respect to leaf size can be identified by the divergence between these two RMSE curves.

From the graph, we can see that as leaf size increases, the in-sample RMSE increases, which means that the model with a very small leaf size is overfitting. However, as the leaf size continues to grow, the in-sample and out-of-sample RMSE converge, meaning that overfitting diminishes. This convergence is most likely due to the increasing leaf size simplifying the model, leading it to generalize better.

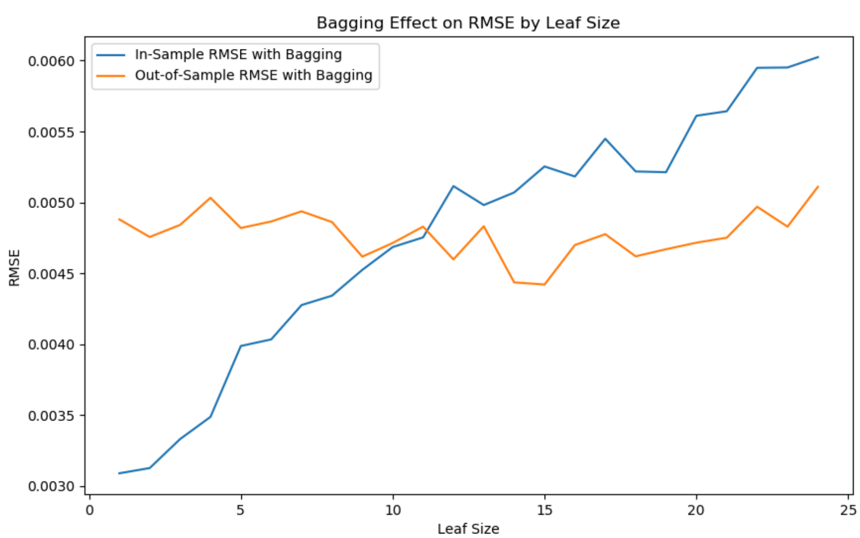
The overfitting starts to occur at very small leaf sizes and continues up to a point where the leaf size is around 5. Beyond this leaf size, the RMSE for both in-sample and out-of-sample plateaus, which indicates the model is neither improving nor overfitting; it may, however, be underfitting as it becomes too generalized.

To reduce overfitting, several strategies can be used. One possibility is to increase the model's leaf size, simplifying it to focus on broader patterns rather than details. Furthermore, cross-validation can help verify that the model can handle new data well by testing it on multiple subsets of the data.

Overall, overfitting is a critical aspect in model training to ensure that predictive models will perform well not only on the data they were trained on but also on new data.

2 EXPERIMENT 2: BAGGING EFFECT ON OVERFITTING

Bagging is used to improve the stability and accuracy of machine learning algorithms. It reduces overfitting by training multiple models using different subsets of the training dataset and then combining their predictions. This approach can increase the overall performance by reducing the variance without increasing the bias too much.



The chart provided above compares the in-sample and out-of-sample RMSE of a Decision Tree Learner with bagging applied, across various leaf sizes using the Istanbul.csv dataset.

In the chart, bagging has a noticeable effect on the RMSE values. The in-sample RMSE with bagging shows an increasing trend as the leaf size grows, which makes sense because larger leaf sizes tend to produce simpler models that do not capture all the unnecessary noise of the training data. Moreover, the out-of-sample RMSE with bagging remains relatively stable across different leaf sizes, indicating that bagging is effective in reducing the gap between training and testing performance, there is less overfitting.

The chart shows that bagging reduces overfitting with respect to leaf size. While the in-sample RMSE increases with leaf size, the out-of-sample RMSE does not increase. Bagging helps the model stay good at making predictions on new, different data, even when we simplify the model by making its decision-making rules less detailed.

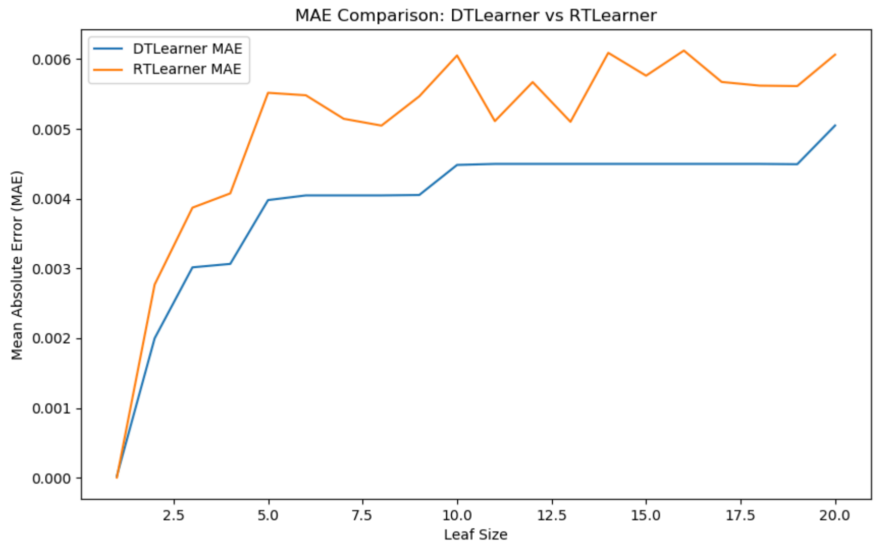
However, it's important to note that while bagging seems to reduce overfitting, it does not eliminate it completely. The out-of-sample RMSE with bagging does show some fluctuations and a slight upward trend as leaf size increases, which could indicate a small degree of overfitting or possibly the start of underfitting.

In conclusion, bagging can reduce overfitting in decision tree learners with respect to leaf size. It accomplishes this by averaging the predictions from multiple models, each trained on different subsets of the data, which helps to cancel out the errors from any one overly complex model. While bagging can help the model, it's not a fix-all solution. It should be used carefully along with other methods to make sure the model isn't too complicated and can still make good predictions on new data it hasn't seen before.

3 EXPERIMENT 3: COMPARISON OF DTLEARNER AND RTLEARNER

In Experiment 3, we quantitatively compared the "classic" decision trees (DTLearner) against random trees (RTLearner) using two different metrics:

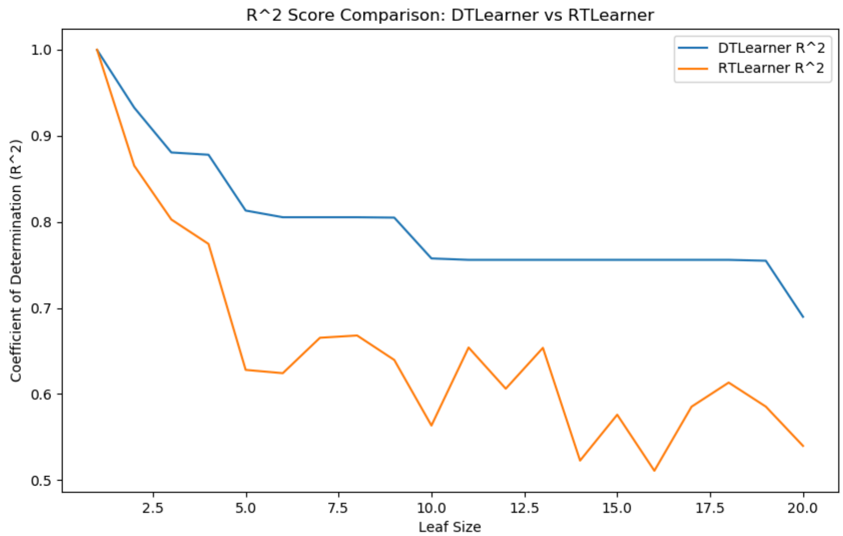
Mean Absolute Error (MAE) and Coefficient of Determination (R^2), as depicted in the provided charts.



3.1 MAE Comparison

The Mean Absolute Error (MAE) represents the average magnitude of the errors in a set of predictions, without considering their direction. It's a measure of accuracy that summarizes the difference between predicted and actual values.

From the MAE comparison chart, the DTLearner shows an overall lower MAE across most leaf sizes compared to the RTLearner, suggesting that the DTLearner generally makes more accurate predictions. The RTLearner's MAE is consistently higher, indicating less accuracy in its predictions. This could be due to the random nature of the RTLearner, which may lead to less consistent predictions.



3.2 R^2 Score Comparison

The R^2 score tells us how well the model's predictions match the actual data. A score of 1 means perfect predictions, and a score of 0 means the model can't predict the data's ups and downs at all.

The chart comparing R^2 scores shows the DTLearner does better with smaller leaf sizes, but its performance drops when the leaf sizes get bigger. The RTLearner's R^2 score changes more and is usually lower than the DTLearner's. The fact that the DTLearner does better with smaller leaf sizes means it's better at matching the data's changes in those cases. This could be because it uses a more organized method to decide how to split data, unlike the RTLearner, which uses more random methods.

3.3 Performance and Superiority

Based on the tests, the DTLearner did better because it was more accurate (lower error) and better at predicting changes (higher R^2) when using smaller leaf sizes. This better result might come from how traditional decision trees work. They predict better when the right amount of detail (leaf size) is used.

The RTLearner's results show it could handle changes in leaf size better, with smaller shifts in performance across different leaf sizes. This means that in situations where it's hard to know the best leaf size or when the data changes a lot, the RTLearner might do better than the DTLearner. This is because its random nature acts like a stabilizer.

No learner is likely to always be superior to the other. The DTLearner may perform better in scenarios where the relationship between features and the target variable is well-understood and can be modeled with clear decision boundaries. In contrast, the RTLearner may offer advantages in scenarios with complex interactions and high variability, where overfitting is a significant concern. The choice between the two should be based on the specific dataset characteristics and the problem's context.

Summary

The findings indicate a complex relationship between leaf size and model overfitting, with bagging presenting a possible solution for its mitigation. The comparative study revealed that while DTLearner may offer more accuracy in some cases, RTLearner could provide more consistency across different scenarios. Future investigations could explore other ensemble methods and their effects on decision tree learners' performance.